Demonstration of Tests of Conversion of PM Dot Notation to Parentheses

Dennis J. Darland

October 5, 2025

SECTION 0. VERIFICATION TESTS (of dot to paren dot icn)
For each proposition is given:
1: the PM notation with dots.
2: the notation with parentheses
3: the Polish (with Lukasiewicz symbols) notation

$\ast 2{\cdot}06 \vdash \therefore p \supset q \,.\supset : q \supset r \,.\supset .\, p \supset r$
Version with parentheses
$\ast 2{\cdot}06 \vdash (p \supset q) \supset ((q \supset r) \supset (p \supset r))$
Polish Lukasiewicz notation
$\ast 2{\cdot}06 \vdash CCpqCCqrCpr$
$\ast 3{\cdot}47 \vdash \therefore p \supset r \,.\, q \supset s \,.\supset : p \,.\, q \,.\supset .\, r \,.\, s$
Version with parentheses
$\ast 3{\cdot}47 \vdash (p \supset r) \wedge (q \supset s) \supset ((p) \wedge (q) \supset (r) \wedge (s))$
Polish Lukasiewicz notation
$\ast 3{\cdot}47 \vdash CKCprCqsKCKpqrs$
$\ast 4{\cdot}22 \vdash : p \equiv q \,.\, q \equiv r \,.\supset .\, p \equiv r$
Version with parentheses
$\ast 4{\cdot}22 \vdash (p \equiv q) \wedge (q \equiv r) \supset (p \equiv r)$
Polish Lukasiewicz notation
$\ast 4{\cdot}22 \vdash CKEpqEqrEpr$
$\ast 4{\cdot}41 \vdash \therefore p \,.\vee .\, q \,.\, r : \equiv .\, p \vee q \,.\, p \vee r$
Version with parentheses
$\ast 4{\cdot}41 \vdash ((p) \vee (q) \wedge (r)) \equiv (p \vee q) \wedge (p \vee r)$

Polish Lukasiewicz notation
$\ast 4{\cdot}41 \vdash KEKApqrApqApr$
$\ast 4{\cdot}43 \vdash :. p . \equiv : p \vee q . p \vee \sim q$
Version with parentheses
$\ast 4{\cdot}43 \vdash (p) \equiv ((p \vee q) \wedge (p \vee \sim q))$
Polish Lukasiewicz notation
$\ast 4{\cdot}43 \vdash EpKApqApNq$
$\ast 4{\cdot}44 \vdash :. p . \equiv : p . \vee . p . q$
Version with parentheses
$\ast 4{\cdot}44 \vdash (p) \equiv ((p) \vee (p) \wedge (q))$
Polish Lukasiewicz notation
$\ast 4{\cdot}44 \vdash EpKAppq$
$\ast 4{\cdot}87 \vdash :. p . q . \supset . r : \equiv : p . \supset . q \supset r : \equiv : q . \supset . p \supset r : \equiv : q . p . \supset . r$
Version with parentheses
$\ast 4{\cdot}87 \vdash ((p) \wedge (q) \supset (r)) \equiv ((p) \supset (q \supset r)) \equiv ((q) \supset (p \supset r)) \equiv ((q) \wedge (p) \supset (r))$
Polish Lukasiewicz notation
$\ast 4{\cdot}87 \vdash EEECKpqrCpCqrCqCprCKqpr$
$\ast 4{\cdot}88 \vdash :. p . q . \supset . r . \equiv : p . \supset . q \supset r : \equiv : q . \supset . p \supset r : \equiv : q . p . \supset . r$
Version with parentheses
$\ast 4{\cdot}88 \vdash (p) \wedge (q) \supset (r) \equiv ((p) \supset (q \supset r)) \equiv ((q) \supset (p \supset r)) \equiv ((q) \wedge (p) \supset (r))$
Polish Lukasiewicz notation
$\ast 4{\cdot}88 \vdash EEECKpqrCpCqrCqCprCKqpr$
$\ast 5{\cdot}33 \vdash :. p . q \supset r . \equiv : p : p . q . \supset . r$
Version with parentheses
$\ast 5{\cdot}33 \vdash (p) \wedge (q \supset r) \equiv (p) \wedge ((p) \wedge (q) \supset (r))$
Polish Lukasiewicz notation
$\ast 5{\cdot}33 \vdash KEKpCqrpCKpqr$
From Landon D. C. Elkind's Paper in Russell: Vol. 43, no. 1, page 44
$\ast 431{\cdot}441 \vdash . p \vee q . \equiv . r \supset s$
Version with parentheses
$\ast 431{\cdot}441 \vdash (p \vee q) \equiv (r \supset s)$
Polish Lukasiewicz notation
$\ast 431{\cdot}441 \vdash EApqCrs$
$\ast 431{\cdot}442 \vdash : p . \vee . q \equiv r : \supset : s$
Version with parentheses
$\ast 431{\cdot}442 \vdash ((p) \vee (q \equiv r)) \supset ((s))$
Polish Lukasiewicz notation
$\ast 431{\cdot}442 \vdash CApEqrs$

$\ast$431·443$\vdash\!\mathbf{:}\, p \vee q \,\mathbf{.}\!\equiv\!\mathbf{.}\, r \,\mathbf{:}\!\supset\!\mathbf{:}\, s$
Version with parentheses
$\ast$431·443$\vdash ((p \vee q) \equiv (r)) \supset ((s))$
Polish Lukasiewicz notation
$\ast$431·443$\vdash CEApqrs$
$\ast$431·444$\vdash\!\mathbf{:}\, p \,\mathbf{:}\!\vee\!\mathbf{:}\, q \equiv r \,\mathbf{.}\!\supset\!\mathbf{.}\, s$
Version with parentheses
$\ast$431·444$\vdash (p) \vee ((q \equiv r) \supset (s))$
Polish Lukasiewicz notation
$\ast$431·444$\vdash ApCEqrs$
$\ast$431·445$\vdash\!\mathbf{:}\, p \,\mathbf{:}\!\vee\!\mathbf{:}\, q \,\mathbf{.}\!\equiv\!\mathbf{.}\, r \supset s$
Version with parentheses
$\ast$431·445$\vdash (p) \vee ((q) \equiv (r \supset s))$
Polish Lukasiewicz notation
$\ast$431·445$\vdash ApEqCrs$
From same, page 54
$\ast$431·54$\vdash\!\mathbf{:}\, p \,\mathbf{.}\, q \,\mathbf{:}\, r \,\mathbf{.}\, s \,\mathbf{:}\!\supset\!\mathbf{:}\, p \,\mathbf{.}\, s \,\mathbf{:}\, r \,\mathbf{.}\, q$
Version with parentheses
$\ast$431·54$\vdash ((p) \wedge (q)) \wedge ((r) \wedge (s)) \supset ((p) \wedge (s)) \wedge ((r) \wedge (q))$
Polish Lukasiewicz notation
$\ast$431·54$\vdash KCKKpqKrsKpsKrq$
check longer prop name
Version with parentheses
Polish Lukasiewicz notation
Propositions involving quantifiers
$\ast$9·2$\vdash\!\mathbf{:}\, (x) \,\mathbf{.}\, \psi x \,\mathbf{.}\!\supset\!\mathbf{.}\, \psi y$
Version with parentheses
$\ast$9·2$\vdash (((x))\psi x) \supset (\psi y)$
Polish Lukasiewicz notation
$\ast$9·2$\vdash C(x)\psi x\psi y$
$\ast$9·21$\vdash\!\mathbf{:.}\, (x) \,\mathbf{.}\, \psi x \supset \phi x \,\mathbf{.}\!\supset\!\mathbf{:}\, (x) \,\mathbf{.}\, \psi x \,\mathbf{.}\!\supset\!\mathbf{.}\, (x) \,\mathbf{.}\, \phi x$
Version with parentheses
$\ast$9·21$\vdash (((x))\psi x \supset \phi x) \supset (((x))\psi x) \supset ((x))\phi x$
Polish Lukasiewicz notation
$\ast$9·21$\vdash CCC(x)\psi x\phi x(x)\psi x(x)\phi x$
$\ast$9·22$\vdash\!\mathbf{:.}\, (x) \,\mathbf{.}\, \psi x \supset \phi x \,\mathbf{.}\!\supset\!\mathbf{:}\, (\exists x) \,\mathbf{.}\, \psi x \,\mathbf{.}\!\supset\!\mathbf{.}\, (\exists x) \,\mathbf{.}\, \phi x$
Version with parentheses
$\ast$9·22$\vdash (((x))\psi x \supset \phi x) \supset (((\exists x))\psi x) \supset ((\exists x))\phi x$

Polish Lukasiewicz notation

$*9\cdot22 \vdash CCC(x)\psi x\phi x(\exists x)\psi x(\exists x)\phi x$

$*9\cdot31 \vdash :. (\exists x) . \phi x . \vee . (\exists x) . \phi x : \supset . (\exists x) . \phi x$

Version with parentheses

$*9\cdot31 \vdash (((((\exists x))\phi x) \vee ((\exists x))\phi x)) \supset ((\exists x))\phi x$

Polish Lukasiewicz notation

$*9\cdot31 \vdash CA(\exists x)\phi x(\exists x)\phi x(\exists x)\phi x$

$*9\cdot401 \vdash :: p : \vee : q . \vee . (\exists x) . \psi x :. \supset :. q : \vee : p . \vee . (\exists x) . \psi x$

Version with parentheses

$*9\cdot401 \vdash (((p) \vee ((q) \vee ((\exists x))\psi x))) \supset ((q) \vee ((p) \vee ((\exists x))\psi x))$

Polish Lukasiewicz notation

$*9\cdot401 \vdash CApAq(\exists x)\psi xAqAp(\exists x)\psi x$

$*10\cdot35 \vdash :. (\exists x) . p . \psi x . \equiv : p : (\exists x) . \psi x$

Version with parentheses

$*10\cdot35 \vdash (((\exists x))p) \wedge (\psi x) \equiv (p) \wedge (((\exists x))\psi x)$

Polish Lukasiewicz notation

$*10\cdot35 \vdash KEK(\exists x)p\psi xp(\exists x)\psi x$

$*11\cdot2 \vdash : (x, y) . \phi[x, y] . \equiv . (y, x) . \phi[x, y]$

Version with parentheses

$*11\cdot2 \vdash ((x, y) . \phi[x, y]) \equiv ((y, x) . \phi[x, y])$

Polish Lukasiewicz notation

$*11\cdot2 \vdash E(x, y) . \phi[x, y](y, x) . \phi[x, y]$

One Step in proof of 11.55 I wanted example of 2 adjacent quantifiers - hard to find.

$*11\cdot551 \vdash :. (x) :. (\exists y) . \psi x . \phi[x, y] . \equiv : \psi x : (\exists y) . \phi[x, y]$

Version with parentheses

$*11\cdot551 \vdash ((((x))(\exists y))\psi x) \wedge (\phi[x, y]) \equiv (\psi x) \wedge (((\exists y))\phi[x, y])$

Polish Lukasiewicz notation

$*11\cdot551 \vdash KEK(x)(\exists y)\psi x\phi[x, y]\psi x(\exists y)\phi[x, y]$

From same, page 46

$*431\cdot46 \vdash : (x) . \psi x . \phi x . \supset . (x) . \psi x$

Version with parentheses

$*431\cdot46 \vdash (((x))\psi x) \wedge (\phi x) \supset ((x))\psi x$

Polish Lukasiewicz notation

$*431\cdot46 \vdash CK(x)\psi x\phi x(x)\psi x$

Other Tests

$*99\cdot99 \vdash :. \sim(\exists x) : \sim\psi x . \supset . (x) . \sim\psi x$

Version with parentheses

$*99{\cdot}99 \vdash ((\sim(\exists x))\sim\psi x) \supset ((x))\sim\psi x$
Polish Lukasiewicz notation
$*99{\cdot}99 \vdash CN(\exists x)N\psi x(x)N\psi x$